

Business Analytics and Big Data MSc 2024-25

Details

Programme Title	Business Analytics and Big Data
Short Title	Business Analytics and Big Data
Programme Code	MSC/BABD
Academic Year	2024-25
Org Unit	ULMS PG
Other Org Units	n/a
School/Institute	Management School
Faculty	Faculty of Humanities and Social Sciences
Awarding Institution	University of Liverpool
Teaching Institution(s)	University of Liverpool
Partner Institution(s)	n/a
Delivery Location	Liverpool campus
Mode of Delivery	Full-time
Duration	1 Years
Entry point (intake)	September
Status	Validated
Final Award	Master of Science (MSc)

Programme Leader	Cagatay Iris
Proposal Owner	Sarah Isenberg
Delivered From	23 Sep 2024

Blurb

Overview

The aim of the programme is to provide education at postgraduate level in the theories and applications in the area of Business Analytics and Big Data and to jointly prepare tomorrow's managers and business executives with unique exposure to current business opportunities, challenges and developments in the area of Business Analytics and Big Data as applied in management.

The programme aims to satisfy the growing demand by the industry and business communities for highly skilled individuals with a sound strategic and applied understanding of the business opportunities and challenges in the rapidly emerging and dynamic field of Business Analytics and the management of Big Data.

Careers

The aim of the programme is to jointly prepare tomorrow's business leaders with unique exposure to the rapidly evolving field of Big Data and the opportunities, challenges and developments associated with running or expanding digital business enterprises.

Through the exposure to strategic initiatives in the management of Big Data, as well as case-based workshops/practical application developments for data mining, analysis and visualisation using industry-leading software, such as Teracota, graduates of the programme shall find themselves well equipped to meet the many challenges associated with strategic, operational and technological adaptation in the digital enterprise.

Students are eligible to avail one-to-one career planning support from the ULMS MSc Careers Adviser.

As well as developing a range of subject-specific and technical skills in Business Analytics and Big Data, students will also gain more general skills to enhance their employability. These include:

knowledge and awareness of business organisations;

numerical and quantitative skills;

problem-solving and analytical ability;

oral and written communication skills;

ability to construct an argument as a basis for building a commercial case and for developing negotiation skills;

knowledge of global business issues and business language skills.

Students on the programme will have access to the university's careers service to enhance employability skills and give access to a wide range of employers. Academics also have good relationships with local employers.

Educational Aims

To promote a systematic understanding of knowledge, and a critical awareness of current problems and new insights, which are at the forefront of Business Analytics and the use of Big Data in business and management;

To support the development of subject specific and key transferable skills necessary for employment associated with the strategic and operational function of Business Analytics and the management of Big Data;

To provide opportunities for students to acquire a range of analytical techniques and approaches required for the effective tactical and strategic management of Big Data within organisations;

To enable students to develop their understanding of data management strategies and practices adopted in Business Analytics and Big Data, and to gain familiarity with relevant data analysis software;

To support the enhancement of skills of written communication.

Teaching and Learning Methods

The learning, teaching and assessment strategy for the programme ensures that the subject-based learning outcomes are delivered and that students have the opportunity to develop the key skills during their year of study.

The strategy draws upon the requirements and recommendations of professional accrediting bodies such as AACSB (The Association to Advance Collegiate Schools of Business). A number of learning goals have been defined for AACSB accreditation purposes and modules are reviewed annually to

ensure that there is evidence of learning in relation to each of these goals. The AACSB goals are embedded in the outcomes and skills sections, above. The AACSB requirement that all learning goals must be assessed is now a formal part of School strategy and any proposed changes to the assessment methods used on a module are scrutinised by the Board of Studies in relation to both the programme's requirements and the AACSB requirements.

The School ensures that students are equipped to meet the learning outcomes of the programme by providing opportunities to develop learning skills throughout the programme. Outside the programme, there are a number of School-wide strategies that ensure that students have opportunities to develop their skills.

Delivery of the programme syllabus is via lectures, practical work and case studies, tutorials and project work. Lectures provide the principal approach to imparting subject knowledge and constitute most of the formal contact time with students. In addition, a concerted effort has been made to include a significant amount of case study material and practical work in order to demonstrate the application of principles and theories delivered within the main lecture programme. The practical elements of the programme take the form of both team-based and individual or self-directed exercises.

The programme has been designed for both practising professionals studying part-time alongside full-time students. Industrial and business experience and prior learning will be quite diverse. The Introduction to Management module has been specifically designed to introduce students to the study of management at postgraduate level taking account of the probable lack of management and business experience of full-time students and recent study experience of part-time. In addition to subject-specific content relating to both contemporary and future management challenges and the alternative e-Business approaches and strategies for addressing these issues, the module will provide the means for students to appreciate the interdisciplinary nature of management. It will also provide the opportunity to work in teams, undertake literature searches and improve study and communication skills.

The emphasis in both delivery and assessment is, necessarily at this level, on the demonstration of competence and application rather than appreciation and knowledge.

Modules are timetabled in such a manner as to evenly spread the students' workload yet choice and availability of elective modules could compromise this plan. Students will be counselled if option choice results in an uneven distribution of work.

Module and programme feedback is gained through the regular use of feedback questionnaires, via the Staff-Student Liaison Committee and through an annual external examiner's report. The Director of Studies is responsible for acting on the feedback and recommendations received.

The School has a team of dedicated Student Learning and Teaching Support Officers (LTSO) who provide additional support outside the curriculum. The role was introduced at the start of the 2011-12 academic year and currently the LTSOs provide drop-in sessions for students requiring academic support (such as help with referencing skills, research etc.) and also provide sign-up academic skills sessions. The LTSOs have also produced a study skills handbook which is provided to all students at the start of the academic year, which is also available for download on the School's website. The LTSO acts

as the School's Disability Officer and works with the University's Disability Support Team to ensure that appropriate academic support is provided to those students who need it.

The School recognises the importance of e-Learning to the curriculum and encourages staff to use this in the individual module learning, teaching and assessment strategies. Support for further development in this area will be provided by a member of the Professional Services Team in the School following the recent appointment of a TEL Technician. The TEL Technician will work with staff to help them to utilise software to its fullest extent for teaching and assessment, as well as identifying other e-Learning opportunities that may be suitable for development within the curriculum.

Assessment Strategy

The School's strategy requires that a diverse range of assessment methods should be used throughout a programme. This ensures that students have a range of assessment opportunities in which to provide evidence of their learning. The benefits of this are twofold. Firstly, there are wider opportunities for students to develop essential learning and employability skills, including teamwork, presentation skills, independent learning and written communication. Secondly, the School recognises the diversity of its student body and that some students may perform better in some types of assessment than others. By offering a range of assessment methods, the School ensures that all students have opportunities to excel as well as opportunities to develop their skills in assessment areas where they feel less confident.

Examinations are of two-hours duration. Certain modules are assessed by examination and coursework, others entirely by coursework. Those modules with a significant coursework component are focused on research in the area of Big data management, the application of Big Data analysis tools and technologies or are case-driven.

Reference Points and Benchmarks n/a

Variations to Regulations n/a

Professional Statement n/a

Admissions

Admission Criteria

Tariff Points

Interview/Audition

Portfolio Required

Misc

Benchmark Subject(s) Business and Management (Master's level) 2023

External Examiner(s)

Foundation Year

Placement or Exchange?

No

Further Details

Sandwich Year

Year Abroad

No

Credit Recognition

Languages of Tuition

Documents Link

UCAS Code

n/a

Related Programmes

n/a

Notes

Finance

Tuition Fee Band

Additional Fee Information

n/a

Narrative

Inclusivity and Diversity

The programme design, structure and content are consistent and compliant with the University's Diversity and Equality of Opportunity Policy.

The programme is delivered entirely at the Liverpool campus, which meets extant accessibility requirements.

The School's Learning and Teaching Support Officers identify and support students with disabilities by providing a bespoke support package for each student, including adaption of the assessment process where necessary. This is communicated to and implemented by all staff involved in supporting the student.

Each student is allocated an Academic Adviser, whose role is to provide academic guidance to the student during the period of the programme. Academic support for each module is available to students both in person and by email from the lecturer of each module.

Diversity is a key objective. It is planned by targeting admissions from multiple international markets and the home market. A limited number of scholarships are also available and these are used to promote diversity.

For students whose first language is not English, English language support is available. Regular English language classes are available during the academic year. In consultation with the academic staff, these can be tailored specifically to the programme, using practical scenarios that are realistic to the register, level, length of scenarios and complexity of English necessary for successful completion of the programme.

Widening participation is key the university's practices, policies and procedures and the University of Liverpool is one of the most successful in the Russell Group in achieving widening participation by accepted measures.

The University encourages equality of opportunity and Management School programmes have achieved a good gender balance.

Costs						
Cost Type	Description	Value Type	Cost	Minimum Cost	Maximum Cost	Pathways
Student Cost	Printing and calculator	Approximate	43.00			

Awards						
Title	Year of Delivery	Award Type	Level	Award Title (if different from the Programme Title)	Pathways	
MSc Business Analytics and Big Data	2024-25	Target	Level 7			
PGCert Business Analytics and Big Data	2024-25	Exit	Level 7			
PGDip Business Analytics and Big Data	2024-25	Exit	Level 7			
UoL PGA Unnamed	2024-25	Exit	Level 7	Unnamed		

Subject(s)	
HECoS Subject	Proportion
100755: Data Management	100.0 %

Accreditation							
PSRB	Type	Status	Description	Start Date	End Date	Review Date	Pathways
Association to Advance Collegiate Schools of Business (AACSB)	Accredited	Active					
European Quality Improvement System (EQUIS)	Accredited	Active					

Pathways		
Title	Educational Aims	Pathway Leader

Outcomes			
Category	No. / Ref	Learning Outcome / Skills	Pathways
Learning Outcomes: Master's	1 / M1	Students will have a critical awareness of the role of Business Analytics and Big Data as well as the impact of contemporary IT tools and digital technologies on various industry sectors.	
Learning Outcomes: Master's	2 / M2	Students will be able to critically appraise the role of Big Data for enabling business growth and creating/supporting opportunities in the areas of marketing and product development.	
Learning Outcomes: Master's	3 / M3	Students will be able to formulate and develop an integrated corporate strategy for the management of Business Analytics and Big Data and evaluate the impact of contemporary IT tools and digital technologies within organisations.	
Learning Outcomes: Master's	4 / M4	Students will be able to demonstrate an in-depth knowledge of big data analytics and the use of statistical processing tools and techniques.	
Learning Outcomes: Master's	5 / M5	Students will be able to discuss key technology trends in Business Analytics, in-memory computing and data mining.	
Learning Outcomes: Master's	6 / M6	Students will be able to assess opportunities and limitations associated with large-scale data management and will have a critical awareness of Internet security policies relating to data retrieval and storage, privacy and ethical considerations.	
Learning Outcomes: Master's	7 / M7	Students will be able to develop strategic management and business growth policies.	
Learning Outcomes: PGDip	1 / PGDip1	Students will have a critical awareness of the role of Business Analytics and the impact of Big Data on various industry sectors.	
Learning Outcomes: PGDip	2 / PGDip2	Students will be able to critically appraise the role of Business Analytics and Big Data and related contemporary IT tools and digital technologies for enabling business growth and creating/supporting opportunities in the areas of marketing and product development.	
Learning Outcomes: PGDip	3 / PGDip3	Students will be able to formulate and develop an integrated corporate strategy for the management of Business Analytics and Big Data and evaluate the impact of contemporary IT tools and digital technologies within organisations.	
Learning Outcomes: PGDip	4 / PGDip4	Students will be able to demonstrate an in-depth knowledge of big data analytics and have basic knowledge of statistical processing tools and techniques.	
Learning Outcomes: PGDip	5 / PGDip5	Students will be able to discuss key technology trends in Business Analytics, in-memory computing and data mining.	
Learning Outcomes: PGDip	6 / PGDip6	Students will have a critical awareness of Internet security policies relating to data retrieval and storage, privacy and ethical considerations.	

Outcomes			
Category	No. / Ref	Learning Outcome / Skills	Pathways
Learning Outcomes: PGCert	1 / PGCert1	Students will be able to demonstrate knowledge of the role of Business Analytics and Big Data as well as and the impact of contemporary IT tools and digital technologies on various industry sectors.	
Learning Outcomes: PGCert	2 / PGCert2	Students will be able to appraise the role of Business Analytics and Big Data and related contemporary IT tools and digital technologies for enabling business growth and creating/supporting opportunities in the areas of marketing and product development.	
Learning Outcomes: PGCert	3 / PGCert3	Students will be able to develop an integrated corporate strategy for the management of Business Analytics and Big Data and consider the impact of contemporary IT tools and digital technologies within organisations.	
Learning Outcomes: PGCert	4 / PGCert4	Students will be able to demonstrate knowledge of Internet security policies relating to data retrieval and storage, privacy and ethical considerations.	
Learning Outcomes: PG Award	1 / PGA1	Students will be able to demonstrate an awareness of the role of Business Analytics and Big Data and the impact of contemporary IT tools and digital technologies on various industry sectors.	
Skills: Learning and Employability	1 / LE1	Communication Skills: Students will have opportunities to develop written communication skills through group discussions and coursework. This will be assessed by written assignments, individual work and the dissertation.	
Skills: Learning and Employability	2 / LE2	Group Working Skills: Some assessed work will require students to work in groups and manage the interaction and relationships with other group members. In doing so, they will gain experience in negotiation, persuasion, influencing and managing conflict.	
Skills: Learning and Employability	3 / LE3	Time management. This applies to all modules of the programme and is relevant for planning scheduled work and meeting assessment deadlines. This will be evident in the students' independent management of their assignments and coursework and by meeting coursework deadlines.	
Skills: Learning and Research	1 / LR1	Problem Solving and Decision Making: Students will be challenged to think critically about business development issues and dilemmas. They will do this by gathering and synthesising information, analysing alternative perspectives and options and presenting a considered opinion or course of action in their programme assessment.	
Skills: Learning, Research and Employability	1 / LRE1	Numeracy Skills: Students will learn financial and statistical techniques. These are developed as part of the modules covering finance and research methods. Students will have the opportunity to analyse and use financial and statistical data to support business decisions through their course work and on appropriate assignments.	

Outcomes			
Category	No. / Ref	Learning Outcome / Skills	Pathways
Skills: Learning, Research and Employability	2 / LRE2	Information and Communication Technology Skills: Students will have opportunities to improve their ICT skills. Students will demonstrate skills in the use of software applications including word processing, visual presentations, data bases, spreadsheets and using the internet for information searches in the course of researching and presenting coursework. They will also have an opportunity to work with specialist data analysis, modelling and simulation tools.	
Skills: Learning, Research and Employability	3 / LRE3	Personal Effectiveness: This applies to all modules where students will be encouraged to think critically about theory and practice and to challenge and critique assumptions about the management of people and organisations. In doing so they will be able to recognise and consider issues such as ethical dilemmas, corporate social responsibility, organisational values, leadership approaches and styles. This will be assessed through attaining the appropriate level of critical engagement with theory and practice in programme assignments.	

Curriculum Map

	LE1	LE2	LE3	LR1	LRE1	LRE2	LRE3	M1	M2	M3	M4	M5	M6	M7	PGA1	PGCert1	PGCert2	PGCert3	PGCert4	PGDip1
Big Data Analytics for Business - EBUS633	✓	✓	✓	✓		✓	✓				✓	✓								
Big Data Management - EBUS622	✓		✓	✓			✓	✓	✓				✓		✓	✓	✓		✓	✓
Data Mining and Machine Learning - EBUS537	✓		✓			✓	✓					✓								
Digital Business Technology and Management - EBUS603	✓	✓	✓	✓		✓	✓	✓	✓	✓					✓	✓	✓	✓		✓
Digital Strategy - EBUS605	✓		✓	✓	✓		✓	✓		✓			✓	✓	✓	✓		✓	✓	✓
AI Ethics and Regulations - EBUS638	✓	✓	✓	✓		✓	✓	✓		✓			✓	✓	✓	✓		✓	✓	✓
BUSINESS SIMULATION AND ANALYSIS - EBUS635	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓		✓	✓	✓	✓		✓
ENTERPRISE SYSTEMS WITH SAP - EBUS612	✓		✓				✓	✓	✓	✓	✓				✓	✓	✓	✓		✓
Global Corporate Strategy - ULMS519	✓		✓		✓	✓	✓			✓				✓				✓		
Global Maritime Logistics - EBUS632	✓		✓				✓				✓									
LOGISTICS AND PHYSICAL DISTRIBUTION - EBUS609	✓		✓				✓							✓						
MSC PROJECT - EBUS621	✓	✓	✓	✓	✓	✓	✓			✓	✓			✓				✓		
OPERATIONS MODELLING AND SIMULATION - EBUS504	✓		✓		✓	✓	✓				✓									
PROJECT AND PORTFOLIO MANAGEMENT IN ORGANISATIONS - EBUS614	✓	✓	✓				✓				✓		✓	✓				✓		
SERVICE OPERATIONS MANAGEMENT - EBUS634	✓		✓				✓							✓						
Sustainable Supply Chain Management - EBUS636	✓		✓	✓			✓			✓			✓	✓				✓		

	PGDip2	PGDip3	PGDip4	PGDip5	PGDip6
 Big Data Analytics for Business - EBUS633			✓	✓	
 Big Data Management - EBUS622	✓				✓
 Data Mining and Machine Learning - EBUS537				✓	
 Digital Business Technology and Management - EBUS603	✓	✓			
 Digital Strategy - EBUS605		✓			✓
 AI Ethics and Regulations - EBUS638	✓	✓			✓
 BUSINESS SIMULATION AND ANALYSIS - EBUS635	✓	✓	✓		
 ENTERPRISE SYSTEMS WITH SAP - EBUS612	✓	✓			
 Global Corporate Strategy - ULMS519		✓			
 Global Maritime Logistics - EBUS632					
 LOGISTICS AND PHYSICAL DISTRIBUTION - EBUS609					
 MSC PROJECT - EBUS621		✓	✓		
 OPERATIONS MODELLING AND SIMULATION - EBUS504			✓		
 PROJECT AND PORTFOLIO MANAGEMENT IN ORGANISATIONS - EBUS614		✓	✓		
 SERVICE OPERATIONS MANAGEMENT - EBUS634					
 Sustainable Supply Chain Management - EBUS636		✓			

Structure					
Semester 1					
Students are required to select two optional modules.					
Code	Module	Credit	Level	Type	Pathway(s)
EBUS622	Big Data Management 2024-25	15	Level 7	Required	
EBUS537	Data Mining and Machine Learning 2024-25	15	Level 7	Required	
EBUS603	Digital Business Technology and Management 2024-25	15	Level 7	Required	
EBUS632	Global Maritime Logistics 2024-25	15	Level 7	Optional	
EBUS504	OPERATIONS MODELLING AND SIMULATION 2024-25	15	Level 7	Optional	
EBUS614	PROJECT AND PORTFOLIO MANAGEMENT IN ORGANISATIONS 2024-25	15	Level 7	Optional	
EBUS634	SERVICE OPERATIONS MANAGEMENT 2024-25	15	Level 7	Optional	

Structure					
Semester 2					
Students are required to select one optional module.					
Code	Module	Credit	Level	Type	Pathway(s)
EBUS633	Big Data Analytics for Business 2024-25	15	Level 7	Required	
EBUS605	Digital Strategy 2024-25	15	Level 7	Required	
EBUS638	AI Ethics and Regulations 2024-25	15	Level 7	Optional	
EBUS612	ENTERPRISE SYSTEMS WITH SAP 2024-25	15	Level 7	Optional	
ULMS519	Global Corporate Strategy 2024-25	15	Level 7	Optional	
EBUS609	LOGISTICS AND PHYSICAL DISTRIBUTION 2024-25	15	Level 7	Optional	
EBUS636	Sustainable Supply Chain Management 2024-25	15	Level 7	Optional	
Semester 3					
Students are required to choose one optional module.					
Code	Module	Credit	Level	Type	Pathway(s)
EBUS635	BUSINESS SIMULATION AND ANALYSIS 2024-25	60	Level 7	Optional	
EBUS621	MSC PROJECT 2024-25	60	Level 7	Optional	

Reporting	
Closed Programme	No
Closed Programme Details	n/a
General Qualification Aim	Master's degree
HESA Cost Centres	Business & management studies